

Multiple Choice (10 marks)

1. Which vitamin is a major lipid-soluble antioxidant in cell membranes?
 - (a) Vitamin A
 - (b) Vitamin D
 - (c) Vitamin E
 - (d) Vitamin K
2. What is the definition of overweight?
 - (a) $\text{BMI} > 25 \text{ kg/m}^2$
 - (b) $\text{BMI} = 25 \text{ kg/m}^2$
 - (c) $\text{BMI} 25\text{-}29.9 \text{ kg/m}^2$
 - (d) $\text{BMI} 25\text{-}30 \text{ kg/m}^2$
3. Current recommendations are that:
 - (a) All babies should be exclusively breastfed until around 6 months
 - (b) Solids should be introduced at 4 months of age
 - (c) Cow's milk is appropriate for all children from 6 months
 - (d) Bottle feeding is more appropriate for babies than breast feeding
4. It is correct to state that for the treatment of Bulimia Nervosa:
 - (a) Fluoxetine is considered of benefit
 - (b) The first approach is to discuss body image
 - (c) It is not useful to discuss weight with patients
 - (d) Treatments should target exclusively on purging practices
5. At what stage(s) during pregnancy did in utero exposure to the Dutch Hunger Winter have the biggest effects on DNA methylation in blood cells from the adult offspring?
 - (a) First trimester of pregnancy.
 - (b) Second trimester of pregnancy.
 - (c) Third trimester of pregnancy.
 - (d) Effects were similar at all stages of pregnancy.

True / False (10 marks)

Answer True or False

6. True or False: A prospective study is the most reliable observational study for exploring the relationship between diet and cancer.
7. True or False: The food-price spike of 2007-2008 was caused by long-term neglect of agricultural development, biofuel production, and commodity speculation.
8. True or False: Infants with severe combined immunodeficiency typically have a strong immune response to common infections and vaccinations.
9. True or False: Low intake of antioxidant nutrients has been linked to the oedema of kwashiorkor in recent studies.
10. True or False: The WHO recommends that individuals should limit their intake of free sugars to more than 10% of their total daily energy intake.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how genetically modified (GM) foods can contribute to global food security and address climate change, including their impact on food supply and nutritional quality.
12. Discuss the factors that have contributed to China's rapid nutrition transition over the past two decades and analyze its impacts on public health and society.

End of Paper